Lab 6

Sequential Circuit Design for Railroad Crossing Signal

1. Introduction

In this lab you will simulate a railroad crossing signal control system. To achieve this goal, you will need to design a finite state machine (FSM) that represents the behavior of the system and then implement the sequential circuit for that FSM . You will use push buttons to represent the railroad sensors and LEDs to represent the signal lights. A switch will be utilized to reset the system to an initial state.

1.2 Railroad Crossing Signal System Requirements

The railroad crossing signal system that we need to simulate is represented in Figure 1. The system has the following specifications and requirements:

a. The railroad is one way only, from east to west.

b. As shown, SE and SW represent the east and west sensors respectively. A sensor outputs logic 1 only when a train is passing by it.

c. The trains can be of any length relative to the distance between SE and SW as shown.

d. No back-to-back trains are allowed (i.e., no train is allowed to pass by SE if the rear of the train ahead of it has not passed away from SW yet).

e. SE and SW are assumed to work perfectly with no wrong output, due to debouncing for example.

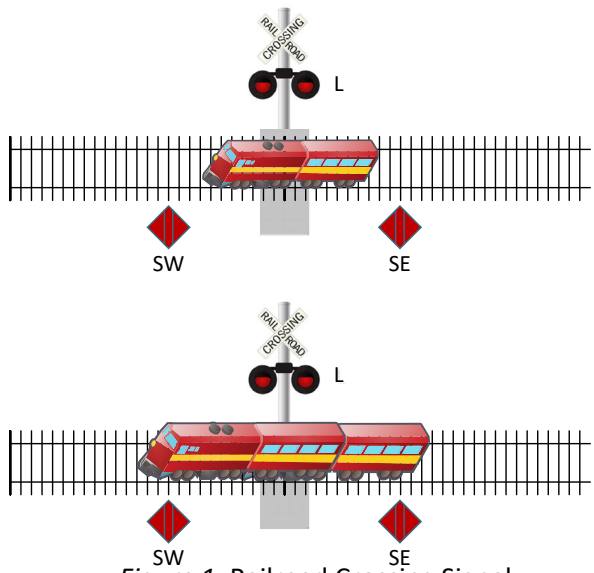

Figure 1: Railroad Crossing Signal

Based on the above requirements, the system can be in any of the following **five states**:

(0) Both SW and SE are off, and the signal light L is also off.
This is the "initial" state of the system, when it is reset.
It is also the "idle" state when there is no train in the crossing area.
- While in this state, it is **irregular** for SW to turn on before SE. This is because no trains from the other direction are allowed in this system.

(1) SW is off, SE is on, and hence L is on. This state happens when a train is passing by SE but has not arrived at SW yet.

(2) Both SW and SE are off while L is on. This state happens when a short train is in the area between SE and SW.
- It is **irregular** while in this state for SE to turn on. This is because back-to-back trains are not allowed.

(3) SW is on, SE is off, and L is on. This state happens when a short train arrives at SW.
- It is **irregular** while in this state for SE to turn on, for the same reason as in state 3.

(4) SW is on, SE is on, and L is on. This state happens when a long train is simultaneously passing by both sensors .
- It is **irregular** while in this state for SE to stay on while SW turns off.

The behavior of the system for both states 3 and 4 (in terms of their expected next states) is identical. So, in the finite state diagram, you need to combine these two states in one state. Therefore, your finite state diagram will have only four states, which will save you one extra flip-flop (why?).

When we build the transition table of a sequential circuit we usually deal with irregular inputs as "don't care". However, as the railroad crossing signal is a sensitive system, we cannot allow the system to be in an unknown state even when an irregular input is provided by mistake. Therefore, in your state diagram, assume that when irregular inputs are received, the trains are halted until this irregularity is fixed. Therefore, handle irregularities by keeping the system in its most recent state.

Pre-Lab Assignment
Based on the previous description of our railroad crossing signal system, design the finite state diagram for the system. The states have been already explained, however, you need to come up with the transitions between these states that are triggered by the different input combinations from the SE and SW sensors.

Have the prelab assignment answers ready to be checked at the beginning of the lab session.

2. Sequential Circuit Implementation

Based on your finite state diagram from the pre-lab assignment, design the circuit following the steps explained in the lecture. As your FSM will have four states and two inputs, you will need two *D* flip-flops to represent these two states. Along with the **SW** and **SE** inputs, your transition table should look as shown in Table 1. Use your state diagram to fill the $\mathbf{D_A}$ and $\mathbf{D_B}$ columns in the table and utilize two Karnaugh maps to implement their minimized Boolean expression.

Current State		Inputs		Next State	
Q_A	Q_B	SW	SE	D_A	D_B
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	1	0
0	1	0	1	0	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	1	0
1	0	0	1	1	0
1	0	1	0	1	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1
1	1	1	1	1	1

Table 1: FSM transition table

Use the following components in your design:

- The needed *D* flip flops can be found in the **Symbol Tool → Primitives → Storage → dff**.
- **SE** and **SW** must be connected to pushbuttons **KEY[0]** and **KEY[3]** respectively. The connection must be through an inverter as the board pushbuttons are **active low** (i.e., their output is zero when pushed).
- A **reset** input, which resets the flip flops in your circuit to state (0), must be connected to switch **SW[0]**.
- The light signal is represented by **LEDR[0]**.

Hint: For debugging purposes, it is recommended that you add two LEDs to show the flip flops outputs: $\mathbf{Q_A}$ and $\mathbf{Q_B}$.

Assignment 1

Upload your design to the DE1-SoC board and test it using the **SW[0]**, **KEY[0]**, and **KEY[3]**. Test all scenarios of train movements for short and long trains. Also, test your circuit when it receives any irregular input from the push buttons. Record a 30-second (maximum) video testing all these scenarios and upload it with your submission. Include in your report a screenshot of your design.

3. Extra Credit Enhancement (20 points)

Instead of just turning on the LED representing the railroad crossing signal, enhance your design to simulate the behavior of a real railroad crossing signal where two lights are used (as shown in Figure 1). These two lights, when they are active, keep flashing in an alternating fashion. Assuming that the alternating frequency is 1 second. Utilize two LEDs and any needed digital circuits from the counters lab assignment to implement this enhancement.

Extra Credit Assignment

Upload your enhanced design to the DE1-SoC board and take a 30-second (maximum) video showing how it works with different input scenarios. Upload the video with your lab submission on Canvas. Include a screenshot of your enhanced design and explain it in the lab report.

4. Lab Submission Instructions

- The lab is intended to be implemented by each **individual** student. It is also allowed for a **maximum of two students** to work in the lab together. If you decide to work in a group of two students, **both students** must **check off** the working lab assignments with a lab instructor and only **one student** must complete the lab submission as explained in the lab assignment on Canvas.
- Write your lab report following the **report template** provided on Canvas. If you are working with a partner, write both names/IDs on the cover page. It is **prohibited** to include a partner's name on a lab report cover page if that partner does not actively participate in implementing the lab.
- The pre-lab questions prepare you for the challenges you will be presented with in the actual lab. Have their answers ready before the lab session and include these answers in the lab report together with the lab assignments.
- For each lab, submit the following on Canvas before the announced due date/time (only submissions on Canvas are accepted):
    - The lab report as a single Word or PDF document
    - Any files the assignments might ask for (e.g., a demonstration video file)
    - Submit each of the above files separately (do not upload compressed files).
- You can submit multiple attempts for this lab; however, only what you submit in the last attempt will be graded (i.e., all required reports and files must be included in this last attempt).
- **Late submission** of any lab assignment after the given due date and time will be penalized at 1 point per every late hour. No assignment submission will be accepted 48 hours after its due date/time.
- **Successful submission**: it is your responsibility to verify that you submitted successfully all the required files for the lab. First, to verify that your files were submitted, go to the assignment details page, and make sure that it says "Successfully submitted" under grade. Second, you need to validate your submission details by clicking the Submission and Rubric link on that page.
- **Deliberate plagiarism** in any assignment is an extremely serious offense that may result in failing the course. Plagiarism includes submitting answers that are not yours (e.g., copied from someone else or from the Internet). Copying materials from the course slides posted on Canvas is not considered plagiarism.